EXPERIMENT DESIGN DOCUMENT

Prospective A/B test of a service-quality improvement initiative

Version 1.0 | Prepared for the fictional Noida multi-brand car service chain

1. Executive summary

This document proposes a controlled A/B test to estimate whether an enhanced service-quality process reduces subsequent warranty claims without causing unacceptable harm to service revenue, repeat repairs, customer experience, or operational capacity. The design is prospective: eligible vehicles/customers are assigned before the intervention, outcomes are measured over a pre-specified follow-up window, and the primary analysis follows the original assignment (intention-to-treat).

Planning note: the intervention details, baseline rate, minimum detectable effect, eligible volume, and follow-up window below are explicit working assumptions for a project draft. Confirm them against the actual operational proposal and uploaded datasets before treating the design as final.

2. Business question and hypothesis

Business question: Does the enhanced service-quality initiative reduce the probability of a warranty claim for an eligible vehicle during the follow-up period, compared with the existing service process?

Treatment hypothesis (H1): The treatment group has a lower warranty-claim rate than the control group.

Null hypothesis (H0): The treatment and control groups have the same warranty-claim rate.

The intervention should be documented precisely before launch. A practical example is a standardized multi-point quality check, technician sign-off, and proactive maintenance recommendations delivered consistently at the participating service visit. Do not change the intervention midway through the test without recording a protocol amendment.

3. Treatment, control, and eligibility

Treatment group	Control group
Receives the new standardized quality-check process and all defined supporting steps.	Continues the current standard service process; no access to the new treatment during the test.

Eligibility (finalize before randomization):

- Include vehicles/customers meeting the chain's pre-defined service and warranty eligibility rules during the enrollment period.
- Exclude records with invalid or unresolvable vehicle identifiers, ineligible warranty status, or outcomes already occurring before assignment.
- Define one index service event per experimental unit to avoid counting the same vehicle repeatedly as independent observations.
- Record exclusions and reasons before examining outcomes; apply identical rules to both arms.

4. Randomization unit and allocation

Preferred design: randomize at the vehicle/customer level within each dealership, using a reproducible 1:1 allocation and stratifying by dealership (and, if useful, major vehicle brand or service type). This controls for fixed differences between dealerships while keeping treatment and control contemporaneous.

Use vehicle/customer-level assignment only if staff can reliably deliver the assigned process and contamination between groups is unlikely. If the intervention changes the whole dealership's workflow, randomize at dealership level instead. There are only 18 dealerships, so a dealership-randomized design has few clusters and requires cluster-aware power calculations and inference; do not treat every vehicle as an independent randomized unit.

Maintain a permanent assignment table with experiment ID, unit ID, dealership, assignment date, assigned arm, and eligibility status. Analyze by original assignment (intention-to-treat), even if some treatment units do not receive every step.

5. Metrics and guardrails

Metric	Operational definition	Role
Warranty claim rate	Number of eligible randomized vehicles with ≥1 qualifying warranty claim within the follow-up window ÷ eligible randomized vehicles with complete outcome ascertainment.	Primary outcome
Repeat-repair rate	Number of eligible vehicles with a repeat repair for the same/similar issue within the defined repeat window ÷ eligible vehicles with sufficient follow-up.	Secondary / guardrail
Service revenue per visit	Net service revenue for the index visit (state whether parts, labour, tax, discounts and refunds are included) ÷ eligible index visits.	Financial guardrail
Customer complaints	Complaints per 1,000 eligible visits, using a consistent complaint definition.	Customer guardrail
Service duration	Elapsed time from vehicle check-in to ready-for-collection; compare consistently and account for outliers.	Operational guardrail
Cancellation / refund rate	Cancelled or refunded index visits ÷ eligible index visits.	Financial / operational guardrail

Primary metric recommendation: use vehicle-level warranty claim incidence over a fixed follow-up window if warranty claims are reliably linked to the vehicle and claim dates are complete. If claim coding or linkage is too incomplete, do not silently substitute another metric after seeing results; revise the protocol and justify the alternative before the experiment begins.

6. Sample size, power, and duration

Illustrative planning assumptions (replace with estimates from the historical data):

- Control warranty-claim rate: 8.0%.
- Minimum detectable effect (MDE): reduction of 1.0 percentage point, from 8.0% to 7.0%.
- Two-sided significance level: 5%; statistical power: 80%; allocation: 1:1.
- Expected eligible enrollment volume: 500 experimental units per day across the chain.
- Outcome follow-up: 90 days after the index service/assignment date.

Under these assumptions, a standard two-independent-proportions calculation requires approximately 10,879 units per arm (21,758 total) before allowance for missing outcomes. With a 10% allowance, plan for approximately 12,088 per arm (24,176 total). At 500 eligible units per day, enrollment takes about 49 days, followed by the 90-day outcome window; the approximate end-to-end calendar time is 139 days, excluding data cleaning and final analysis.

These figures are illustrative, not a final power result. Recalculate using the observed baseline rate, realistic MDE, actual eligible enrollment rate, outcome loss, and (if dealership-level randomization is used) the intracluster correlation and cluster-size variation. Do not use the individual-level sample size for a cluster-randomized design.

7. Analysis plan

1. Freeze the analysis plan and metric definitions before looking at treatment outcomes.
2. Validate eligibility, unique experimental units, assignment dates, duplicate IDs, missing outcomes, and date ranges.
3. Check sample-ratio mismatch (SRM) against the planned 50:50 allocation; investigate assignment or logging errors before interpreting effects.
4. Assess baseline balance using standardized mean differences for pre-treatment variables such as dealership, model/brand, vehicle age, and prior service history. Balance checks diagnose randomization; they are not a reason to selectively drop observations.
5. Estimate the primary effect as treatment minus control in percentage points, with a 95% confidence interval and p-value. Report group denominators and event counts.
6. Use a two-proportion method or logistic regression for the binary primary outcome. If assignment is clustered, use cluster-robust inference or a cluster-level/randomization-based method appropriate to the number of clusters.
7. Analyze revenue per visit using a mean difference and confidence interval; consider bootstrap intervals if the distribution is highly skewed. State the unit of resampling.
8. Analyze repeat repairs and guardrails using pre-specified definitions and confidence intervals. Treat secondary outcomes as supporting evidence, not substitutes for the primary outcome.
9. Report intention-to-treat as primary. Any per-protocol analysis is secondary and must be clearly labelled.
10. For historical impact, use Difference-in-Differences only with a defensible comparison region and assess pre-treatment parallel trends; include time and region effects and appropriate clustered uncertainty.

8. Data quality and governance

- Part-code changes: build and validate a dated old-code-to-new-code crosswalk; retain original codes and document ambiguous mappings.
- Missing warranty failure codes: quantify missingness by period, dealership, brand, and treatment arm; do not assume missing means no failure.
- Backwards odometer readings: flag impossible decreases, inspect source records, and avoid automatic correction without an auditable rule.
- Vehicle linkage: validate VIN uniqueness and join cardinality across sales, job cards, warranty claims, and experiment assignment.
- Date logic: confirm sale, service, assignment, claim, and follow-up dates are chronologically plausible.
- Keep a data dictionary, cleaning log, exclusion log, and reproducible analysis code.

9. Decision rules

GO: the primary estimate supports a meaningful reduction in warranty claims, the 95% confidence interval is consistent with a beneficial effect, and no important guardrail shows unacceptable harm. Business owners should define the minimum commercially worthwhile effect and guardrail thresholds before launch.

NO-GO: the data provide credible evidence of no worthwhile benefit or meaningful harm on the primary outcome or a critical guardrail. A non-significant p-value alone is not proof of no effect.

EXTEND / INVESTIGATE: the confidence interval remains too wide to distinguish a worthwhile benefit from no benefit or harm, the planned sample/follow-up was not achieved, or material data-quality/SRM/pre-trend issues undermine inference. Do not extend solely because the interim p-value is not significant; any interim monitoring rule must be pre-specified.

10. Threats to causal inference and mitigations

Threat	Mitigation
Contamination	Treatment procedures may spread to control units; separate checklists/training and audit compliance.
Selection / eligibility bias	Apply eligibility rules before assignment and retain an assignment log.
Seasonality and concurrent changes	Run treatment and control concurrently; log promotions, supplier changes, staffing changes, and policy changes.
Outcome misclassification	Standardize claim/repeat-repair definitions and validate vehicle/job-card/claim links.
Missing-not-at-random data	Report missingness by arm and relevant segments; run sensitivity analysis where feasible.
Few clusters	If dealership-randomized, acknowledge only 18 clusters and use small-sample-aware methods; interpret cautiously.
Spillovers and interference	Assess whether treatment for one vehicle changes capacity or service quality for other vehicles.
Multiple comparisons	Pre-specify one primary outcome; label segment and secondary analyses exploratory or adjust for multiplicity.

11. Items to confirm before launch

- Exact intervention steps and implementation owner.
- Final experimental unit: vehicle/customer or dealership.
- Final primary outcome definition and follow-up duration.
- Historical baseline rate, MDE, eligible volume, and outcome completeness.
- Randomization code, assignment storage, and contamination controls.
- Pre-approved guardrail thresholds and decision owner.

Appendix: dataset roles

Vehicle sales can provide VIN, dealership, model, sale date, and vehicle price. Service job cards are needed to define service visits, repairs, labour/parts revenue, and repeat repairs. Warranty claims are needed to identify claim events, claim dates, failure codes, and claim amounts. The experiment dataset must contain the randomization unit, assigned arm, assignment date, and ideally pre-treatment covariates. A historical region/dealership-period panel is needed for the Difference-in-Differences analysis. Confirm actual filenames and column names in the notebook before running the final analysis.